Marcin **Copik**

SERVERLESS · HPC · PHD RESEARCHER

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Summary

I work at the intersection of cloud computing and HPC. My PhD focused on serverless programming models designed to bridge supercomputers and data centers. I developed tailored solutions for different levels of the FaaS computing stack: from hardware and network optimizations to efficient system designs and performance modeling.

Education

PhD in Computer Science
April 2018 - March 2024

ETH ZÜRICH — *Zürich, Switzerland*

- Thesis: High-Performance Serverless for HPC and Clouds
- Advisor: Prof. Torsten Hoefler

Master of Science (MSc) in Simulation Sciences
September 2014 - July 2017

RWTH AACHEN — *Aachen, Germany*

- Grade: 1.5. Interdisciplinary program. Major subject: High-Performance Computing
- Thesis: Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series
- Advisor: Prof. Paolo Bientinesi, Prof. Benjamin Berkels

Summer School in Mathematics
August 2014

UNIVERSITY OF PERUGIA — *Perugia, Italy*

- Courses: Stochastic Processes, Functional Analysis

Bachelor of Science in Engineering (BSc) in Computer Science
September 2010 - March 2014

SILESIAN UNIVERSITY OF TECHNOLOGY — *Gliwice, Poland*

- Grade 5(A). Major subject: Software Engineering
- Thesis: GPU-accelerated stochastic simulator engine for PRISM model checker
- Advisor: Prof. Tadeusz Czachorski
- Additionally, finished two of three years of BSc program in Mathematics (2012-2014).

Experience

Postdoctoral Researcher
Zürich, Switzerland

ETH ZÜRICH — *May 2024 -*

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Assistant
Zürich, Switzerland

ETH ZÜRICH — *April 2018 - April 2024*

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Intern
Redmond, WA, USA

MICROSOFT — *June - October 2019*

- Analyzing microarchitectural implications of serverless workloads.
- Supervisor: Bobbie Manne.

Mentor
STE||AR Group, SPCL.

GOOGLE SUMMER OF CODE — *2017, 2018, 2023, 2024*

- Students: Ajai V George, Gabriel Laberge (co-mentored), Matt Nappo, Abhishek Kumar, Syed Mujtaba, Prajin Khadka.

Student Research Assistant
Aachen, Germany

RWTH AACHEN, HIGH-PERFORMANCE AND AUTOMATIC COMPUTING — *2016 - December 2017*

- Benchmarking linear algebra frameworks.
- Supervisor: Prof. Paolo Bientinesi.

Research Assistant *Baton Rouge, LA, USA*
Louisiana State University, STE||AR Group *April 2016 - August 2016*
- Integrating single-source GPU programming in HPX.
- Supervisor: Prof. Hartmut Kaiser.

Student Research Assistant *Jülich, Germany*
Jülich Supercomputing Centre *October 2014 - March 2016*
- Developing tools for performance analysis of parallel applications at Scalasca.
- Supervisor: Dr Pavel Saviankou.

Software Engineer *Organization: The STE||AR Group*
Google Summer of Code *2015*
- Integrating single-source GPU programming in HPX.
- Supervisor: Dr Hartmut Kaiser.

Software Engineer *Organization: PRISM model checker*
Google Summer of Code *2014*
- Improving statistical model checking.
- Supervisor: Dr Vojtěch Forejt, Dr Dave Parker.

Student Research Assistant *Gliwice, Poland*
The Institute of Theoretical and Applied Informatics *2012 - 2013*
- Implementing GPU simulator of Markov Chains.
- Supervisors: Dr Mateusz Nowak, Dr Artur Rataj.

Student Research Assistant *Gliwice, Poland*
Silesian University of Technology *2012 - 2013*
- Implementing algorithms for registration of respiratory motion.
- Supervisor: Dr Dominik Spinczyk.

Honors & Awards

2025	**SIGHPC Doctoral Dissertation Award**, ACM SIGHPC
2025	**Best Paper Award Nominee**, ACM/IEEE Supercomputing 2025
2025	**Gordon Bell Award in Climate Modeling Finalist**, ACM/IEEE Supercomputing 2025
2024	**Best Research Poster Award**, ACM/IEEE Supercomputing 2024
2023, 2024	**SIGHPC Travel Grant**, awarded for travel to ACM/IEEE Supercomputing 2023 & 2024
2022	**ACM/IEEE George Michael Memorial HPC Fellowship**, awarded for contributions into high-performance serverless
2022	**Gold Medal at the ACM Student Research Competition**, ACM/IEEE Supercomputing 2022
2022	**AWS Cloud Credit for Research Application**
2022	**Google Cloud Research Credits**
2021	**Microsoft Research PhD Fellowship**, awarded for the 2021/2022 academic year
2019	**Gold Medal at the ACM Student Research Competition**, ACM/IEEE Supercomputing 2019

Major Projects

SeBS: Serverless Benchmark Suite *Python, Cloud, FaaS*
FaaS benchmarking suite with automatic build, deployment, and measurements. GitHub: 175 stars, 85 forks. Google Scholar: 231 citations.

rFaaS: RDMA-Enabled FaaS Platform *C++, RDMA, FaaS*
High-performance serverless platform with RDMA acceleration for HPC function invocations. GitHub: 53 stars, 18 forks. Google Scholar: 45 citations.

Peer-reviewed Publications

XaaS Containers: Performance-Portable Representation With Source and IR Containers *Supercomputing*
Copik M., Alnuaimi E., Kamatar A., Hayot-Sasson V., Madonna A., Gamblin T., Chard K., Foster I., Hoefler T. *2025*
- Acceptance Rate 22% (137/623), **Best Paper Award Finalist**

Computing the Full Earth System at 1km Resolution *Supercomputing*
KLOCKE D.. [AND 25 OTHERS, INCLUDING **COPIK M.**] *2025*
- Gordon Bell Award in Climate Modeling Finalist

Core Hours and Carbon Credits: Incentivizing Sustainability in HPC *Supercomputing*
KAMATAR A., GONTHIER M., HAYOT-SASSON V., BAUER A., **COPIK M.**, HOEFLER T., FERNANDEZ R., CHARD K., FOSTER I. *2025*

Regular-Confidential LLM Inference: Performance and Cost Across CPU and GPU TEEs *IISWC*
CHRAPEK M., **COPIK M.**, METTAZ E., HOEFLER T. *2025*
- Acceptance Rate 41.2% (38/91)

DaCe AD: Unifying High-Performance Automatic Differentiation for Machine Learning and Scientific Computing *IEEE Cluster*
BOUDAOUD A., CALOTOIU A., **COPIK M.**, HOEFLER T. *2025*

Cppless: Productive and Performant Serverless Programming in C++ *ACM TACO*
COPIK M., MÖLLER L., CALOTOIU A., HOEFLER T. *2025*

SeBS 2.0: Keeping up with the Clouds *SESAME Workshop @ EuroSys*
COPIK M., CALOTOIU A., HOEFLER T. *2025*

SeBS-Flow: Benchmarking Serverless Cloud Function Workflows *EuroSys*
SCHMID L., **COPIK M.**, CALOTOIU A., BRANDNER L., KOZIOLEK A., HOEFLER T. *2025*
- Acceptance Rate in Fall Round 8.2% (30/367)

A Priori Loop Nest Normalization: Automatic Loop Scheduling in Complex Applications *CGO*
TRÜMPER L., SCHAAD P., ATES B., CALOTOIU A., **COPIK M.**, HOEFLER T. *2025*

Protocol Buffer Deserialization DPU Offloading in the RPC Datapath *IXPUG Workshop @ Supercomputing*
FRANTZ R., GARCIA J., **COPIK M.**, MONROY I., OLMOS J., BLOCH G., DI GIROLAMO S. *2024*

Process-as-a-Service: Unifying Elastic and Stateful Clouds with Serverless Processes *ACM SoCC*
COPIK M., CALOTOIU A., GYORGY R., BÖHRINGER R., BRUNO R., HOEFLER T. *2024*
- Acceptance Rate 30.1% (63/209)

FaaSKeeper: Learning from Building Serverless Services with ZooKeeper as an Example *ACM HPDC*
COPIK M., CALOTOIU A., ZHOU P., TARANOV K., HOEFLER T. *2024*
- Acceptance Rate 17.3% (26/150)

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing *IEEE CiSE*
HOEFLER T., **COPIK M.**, BECKMAN P., JONES A., FOSTER I., PARASHAR M., REED D., TROYER M., SCHULTHESS T., ERNST D., DONGARRA J. *2024*

Software Resource Disaggregation for HPC with Serverless Computing *IEEE IPDPS*
COPIK M., CHRAPEK M., SCHMID L., CALOTOIU A., HOEFLER T. *2024*
- Acceptance Rate 26.1% (88/337)

User-guided Page Merging for Memory Deduplication in Serverless Systems *IEEE Big Data*
QIU W., **COPIK M.**, WANG Y., CALOTOIU A., HOEFLER T. *2023*
- Acceptance Rate 17.5% (92/526)

FMI: Fast and Cheap Message Passing for Serverless Functions *ACM ICS*
COPIK M., BÖHRINGER R., CALOTOIU A., HOEFLER T. *2023*
- Acceptance Rate 29.4% (40/136)

rFaaS: Enabling High Performance Serverless with RDMA and Leases *IPDPS*
COPIK M., TARANOV K., CALOTOIU A., HOEFLER T. *2023*
- Acceptance Rate 25.7% (95/369)

Performance-Detective: Automatic Deduction of Cheap and Accurate Performance Models *ACM ICS*
SCHMID L., **COPIK M.**, CALOTOIU A., WERLE D., REITER A., SELZER M., KOZIOLEK A., HOEFLER T. *2022*
- Acceptance Rate 24.2% (39/161)

MOM: Matrix Operations in MLIR *IMPACT*
CHELINI L., BARTHELS H., BIENTINESI P., **COPIK M.**, GROSSER T., SPAMINATO D. *2022*

Work-stealing Prefix Scan: Addressing Load Imbalance in Large-scale Image Registration *IEEE TPDS*
COPIK M., GROSSER T., HOEFLER T., BIENTINESI P., BERKELS B. *2021*

SeBS: A Serverless Benchmark Suite for Function-as-a-Service Computing *ACM/IFIP Middleware*

COPIK M., KWASNIEWSKI G., BESTA M., PODSTAWSKI M., HOEFLER T. *2021*

- Acceptance Rate 31% (33/107)

Extracting Clean Performance Models from Tainted Programs *ACM PPoPP*

COPIK M., CALOTOIU A., GROSSER T., WICKI N., WOLF F., HOEFLER T. *2021*

- Acceptance Rate 21% (31/150)

GraphMineSuite: Enabling High-Performance and Programmable Graph Mining Algorithms with Set Algebra *VLDB*

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**] *2021*

SISA: Set-Centric Instruction Set Architecture for Graph Mining on Processing-in-Memory Systems *IEEE MICRO*

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**] *2021*

The Generalized Matrix Chain Algorithm *CGO*

BARTHELS H., **COPIK M.**, BIENTINESI P. *2018*

- Acceptance Rate 28.6% (30/105)

Using SYCL as an Implementation Framework for HPX.Compute *DHPCC++ Workshop, IWOCL*

COPIK M., KAISER H. *2017*

A GPGPU-based Simulator for Prism: Statistical Verification of Results of PMC *CS&P*

COPIK M., RATAJ A., WOŹNA-SZCZĘŚNIAK B. *2016*

Methods for abdominal respiratory motion tracking *Computer Aided Surgery*

SPINCZYK D., KARWAN A., **COPIK M.** *2014*

Presentations and Talks

SeBS-Flow: Benchmarking Serverless Cloud Function Workflows *April 2025*

EUROSYS 2025 POSTER

High-Performance Serverless: Bridging the Gap Between Elasticity and Efficiency *March 2025*

INVITED TALK, VHIVE COMMUNITY MEETING.

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing *November 2024*

PANEL MODERATOR AT CANOPIE-HPC WORKSHOP @ ACM/IEEE SUPERCOMPUTING 2024

Serverless HPC: Challenges, Opportunities, and Future Prospects for Accelerated Cloud Computing *November 2024*

PANEL MODERATOR AT ACM/IEEE SUPERCOMPUTING 2024

MIGnificient: Fast, Isolated, and GPU-Enabled Serverless Functions *November 2024*

ACM/IEEE SUPERCOMPUTING 2024 POSTER

Benchmarking Serverless with SeBS: Past, Present, and Future *June 2024*

THIRD INTERNATIONAL WORKSHOP ON SERVERLESS COMPUTING EXPERIENCE 2024

High Performance Serverless for HPC and Cloud *June 2024*

INVITED TALK, INTELLIGENT SERVERLESS AND CLOUD APPLICATIONS SYMPOSIUM, ZURICH UNIVERSITY OF APPLIED SCIENCES.

Evaluating FaaS Systems with the Serverless Benchmark Suite SeBS *June 2024*

SEATED WORKSHOP ON SERVERLESS AT THE EDGE, HPDC 2024

Cppless: Productive and Performant Serverless Programming in C++ *November 2023*

LIGHTNING TALK, LLVM-HPC AT ACM/IEEE SUPERCOMPUTING 2023.

High Performance Serverless for HPC and Clouds *November 2023*

POSTER PRESENTATION AT DOCTORAL SHOWCASE, SC 2023.

Serverless As a Bridge Between HPC and Clouds *May 2023*

INVITED TALK, AWS CLOUD FOR RESEARCH AT ETH.

Serverless As a Bridge Between HPC and Clouds *May 2023*

INVITED TALK, 5TH WORKSHOP ON PARALLEL AI AND SYSTEMS FOR THE EDGE (PAISE), IPDPS 2023.

Serverless As a Bridge Between HPC and Clouds *May 2023*

POSTER PRESENTATION AT PHD FORUM, IPDPS 2023.

Software Resource Disaggregation for HPC with Serverless Computing *November 2022*

ACM/IEEE SUPERCOMPUTING 2022 POSTER, **GOLD MEDAL AT THE ACM STUDENT RESEARCH COMPETITION**.

Software Resource Disaggregation for HPC with Serverless Computing *November 2022*
SuperCompCloud at ACM/IEEE Supercomputing 2022.

Interactive Computing with Serverless Functions in rFaaS *November 2022*
Lightning Talk, UrgentHPC at ACM/IEEE Supercomputing 2022.

Extracting Clean Performance Models from Tainted Programs” *February 2022*
SIAM Conference on Parallel Processing for Scientific Computing (PP22) minisymposium

perf-taint: Taint Analysis for Automatic Many-Parameter Performance Modeling *November 2019*
ACM/IEEE Supercomputing 2019 Poster, **Gold Medal at the ACM Student Research Competition**.

Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series *November 2017*
ACM/IEEE Supercomputing 2017 Poster, ACM Student Research Competition.

HPX and GPU-parallelized STL *May 2016*
C++Now 2016 conference.

Service

Program Committee	EuroSys 2026, SoCC 2025, CANOPIE-HPC 2025, HiPC 2025, IPDPS 2025, EuroSys 2025 (Shadow PC), SESAME 2025
Organizing Committee	PAISE 2025, PAISE 2024
Reviewer	IEEE: TPDS, TC, TCC; Elsevier: Array, Ad-hoc Networks, Journal of Parallel and Distributed Computing, Journal of Systems Architecture, Ocean Modeling; Sage: International Journal of High Performance Computing Applications; LLVM-HPC 2020, ISC 2019
Student Volunteer	ACM/IEEE Supercomputing 2022 & 2023

Teaching

Fall 2024, Fall 2025	Information Systems for Engineers	*ETH Zürich*
Spring 2025	Information Retrieval	*ETH Zürich*
Spring 2024	Parallel Programming	*ETH Zürich*
Fall 2023	Big Data	*ETH Zürich*
Spring 2023	Parallel Programming	*ETH Zürich*
Spring 2022	Parallel Programming	*ETH Zürich*
Fall 2021	Information Systems for Engineers	*ETH Zürich*
Spring 2021	Parallel Programming	*ETH Zürich*
Fall 2020	Compiler Design	*ETH Zürich*
Spring 2020	Parallel Programming	*ETH Zürich*
Fall 2019	Design of Parallel and High-Performance Computing	*ETH Zürich*
Spring 2019	Parallel Programming	*ETH Zürich*
Fall 2018	Numerical Methods for Computational Science and Engineering	*ETH Zürich*

Students

Ryan Zogg	Bachelor Thesis: Performance Modeling and Serverless Adaptation of HPC Workflows	*2025, ETH Zürich*
Eiman Alnuaimi	Master Thesis: Acceleration as a Service (XaaS) Source Containers	*2025, ETH Zürich*
Ritvik Ranjan	Semester Project: Practical Workload Co-location with CRIU and rFaaS	*2025, ETH Zürich*
Simone Kalbermatter	Co-supervised Semester Project: Conducting a base crypto analysis of the SHArP protocol	*2025, ETH Zürich*
Andrea Jiang	Co-supervised Master Thesis: Serverless and Cloud Runtimes for Graph-of-Thoughts	*2024, ETH Zürich*
Entiol Liko	Co-supervised Semester Project: Serverless Co-location with ML	*2024, ETH Zürich*
Oana Rosca	Semester Project: Long-Term Serverless Performance Variability	*2024, ETH Zürich*
Constantin Dragancea	Master Thesis: Adoption and Evolution of C++	*2024, ETH Zürich*

Prajin Khadka	Co-supervised Google Summer of Code Student: Expanded serverless benchmarks	*2024, GSoC*
Syed Mujtaba	Google Summer of Code Student: Using serverless ZooKeeper in Apache projects	*2024, GSoC*
Abhishek Kumar	Co-supervised Google Summer of Code Student: New serverless benchmarks	*2024, GSoC*
Matt Nappo	Co-supervised Google Summer of Code Student: Libfabric Implementation of rFaaS	*2023, GSoC*
Boyan Zhou	Master Thesis: Adoption and evolution of C++ in HPC Applications	*2023, ETH Zürich*
Gyorgy Rethy	Master Thesis: Process-as-a-Service computing on modern serverless platforms	*2022, ETH Zürich*
Laurin Brandner	Master Thesis: Serverless workflows benchmarking	*2022, ETH Zürich*
Lukas Möller	Bachelor Thesis: Serverless C++ Executor	*2022, ETH Zürich*
Malte Wächter	Bachelor Thesis: Profiling and optimizations of serverless functions	*2022, ETH Zürich*
Qiu Wei	Master Thesis: Serverless memory deduplication	*2022, ETH Zürich*
Lukas Tobler	Master Thesis: Serverless GPU functions	*2022, ETH Zürich*
Arnet Colin	Bachelor Thesis: Verification of representativeness of benchmarking suite	*2021, ETH Zürich*
Roman Böhringer	Master Thesis: Serverless collectives.	*2021, ETH Zürich*
Emir İşman	Bachelor Thesis: FaaStest collectives: reliable communication in serverless world	*2021, ETH Zürich*
Konrad Handrick	Co-supervised Bachelor Thesis: Offloading serverless with sPIN	*2021, ETH Zürich*
Tobias Lüscher	Bachelor Thesis: TaintImpact: Taint-Based Change Impact Analysis	*2021, ETH Zürich*
Siegfried Hartogs	Bachelor Thesis: Code-driven Language Development: Framework for Analysis of C/C++ Open-Source Projects	*2021, ETH Zürich*
Lukas Gygi	Bachelor Thesis: CppBuild: Large-Scale, Automatic Build System for Open Source C++ Repositories	*2021, ETH Zürich*
Nicolas Wicki	Bachelor Thesis: Control Flow Taint Analysis for Performance Modeling in LLVM	*2020, ETH Zürich*
Philipp Bomatter	Co-supervised Bachelor Thesis: Towards Extreme-Scale Cache Coherence Protocols and Simulations	*2019, ETH Zürich*
Gabriel Laberge	Co-supervised Google Summer of Code Student: Alternative smart executors	*2018, GSoC*
Ajai V George	Google Summer of Code Student: Work on Parallel Algorithms	*2017, GSoC*